

SKYCREW – Subscription Webapp & Live-Stream Add-on (stretch goals)

Three productized add-ons. All are SaaS subscriptions (the recurring-revenue layer) layered on top of the hardware sale. Comparables: Skybrush Studio (open-source drone-show designer in Blender), Picsui, Lovecast, EventLive, Hitched, Resi (event live-stream platforms, \$19/event to ~\$99/mo), LiveKit (streaming infra).

Part 1 – Venue Planner (the client-facing design webapp)

****Problem it solves:**** couples/clients want to choose their shots, but they can't visualize what "fly in at 2 m, 40° down" means. Photographer pre-visits the venue once, SKYCREW builds a 3D model, client designs the whole camera plan in a browser.

1.1 Photographer side (survey)

1. Operator visits venue (they already do for pre-production – see `00-overview.md §4`).
2. Capture: drone/360° camera sweep + photogrammetry → textured 3D venue mesh ("satellite Google-Maps view" of the room).
3. Upload to webapp → auto-generates walkable 3D model + measure tools (table positions, stage, altar, dance floor).

1.2 Client side (design)

- ****Draw flight paths**** directly on the 3D model (curves, arcs, orbits).
- ****Pick camera angles**** per focal point (altitude, yaw, tilt, lens feel).
- ****Set points of focus**** (head table seat, cake, dance floor) and tie them to trigger pods.
- ****Render preview****: the app renders what each shot will look like (static preview first; live-test footage scrubbed in later versions).
- ****Submit to photographer**** for review – photographer sees every path/angle, flags problems, proposes alternatives the client didn't think of, and client confirms before the event date.

1.3 Tech stack (all open source)

- 3D viewer/editor: Three.js (same pattern as Skybrush Viewer).
- Photogrammetry: open pipeline (OpenSfM/COLMAP or COTS, e.g. RealityCapture-class).
- Backend: Node/Python API + Postgres; object storage for assets.
- Auth/payments: standard SaaS (Stripe, Clerk).
- Hosting: any cloud; media via CDN.

1.4 Pricing

- Included in hardware kits for 12 months; then ****\$29–59/event**** per client project, or ****\$99/mo**** unlimited for Certified Partners (bundled in `03-partner-model.md`).

Part 2 – Collaborative Edit module

****Problem:**** the most-argued-over deliverable is the final edit. Let clients co-decide.

- After the event, footage + per-drone multi-angle sync + pod audio are assembled in the webapp.
- Client can:
 - Review all angles per moment (toast, first dance) and ****star/reject takes****.
 - Pick the hero angle for each beat.
 - Approve a draft highlight cut; request changes inline (timestamped comments).
 - Final-confirm before the photographer exports.

- Photographer keeps creative veto; the flow reduces "revision hell" to one structured pass.
- Pricing: add-on **\$49/event** or included in Partner plans.

Part 3 – Live Streaming + Video Wall

Problem: family that can't travel. Two distinct features:

3.1 Live stream for remote guests

- One private link, any device, no app (Picsui/Lovecast/EventLive pattern – all validate this UX at \$19/event to \$99/mo).
- Source: operator laptop (OBS) ingesting the live drone multiview + a fixed camera; RTMP/WebRTC out; auto-recorded with replay.
- Low latency via LiveKit or Resi-class resilient streaming; adaptive quality per viewer.
- Pricing: **\$19/event** per stream or bundled; unlimited guests; 90-day replay.

3.2 Video wall (bring remote guests INTO the room)

- Optional 43–55" panel(s) at the venue. Layouts the customer can choose:
 - **Grid of streaming guest faces** (their webcams) + virtual **glass-clink to speak** – remote guest clicks the clink, appears with a toast slot on the wall and audio.
 - **Per-table monitors** – a virtual guest is placed at a specific table and can chat with that table's guests (WebRTC room-per-table).
 - **Custom layouts** – however the customer wants it; the framework is a configurable multi-room WebRTC mixer.
- Streaming guests also upload photos/messages to the same gallery (Picsui-style shared real-time gallery for in-person + remote).
- Pricing: hardware panels \$300–1,500 each (see BOM); software **+\$29–99/event** depending on layout complexity.

Part 4 – "Fund-a-Guest": the paid guest-experience menu

Problem it solves: weddings and bar/bat mitzvahs are partly funded by guest gifts, and cash funds (honeymoon, house deposit, tzedakah) already exist as payment rails – but they're disconnected from the film. This turns the event itself into a gift vehicle: **guests pay a fixed price, set by the hosts, to buy a moment** – a song, a talking head, a video-wall message, a karaoke performance. **Hosts enable whichever tiers they want, set the prices, and cap quantities.** It's a menu, not a single feature.

4.1 The menu

#	Tier	What the guest gets	Suggested price
1	Song request	their song plays on the DJ/PA, captioned "requested by [name]"	\$20–60
2	Guest Spotlight	guaranteed 30–90 s drone talking head, always in the final film	\$50–150
3	Video-wall message	host-reviewed recorded message, aired on the video wall	\$100–250
4	Karaoke performance	a karaoke slot on the PA + video wall, in the final film	\$200–500

All paid through the same per-event QR microsite (`\$4.5`) – no app, no login, Apple/Google Pay.

4.2 Tier 1 – Song request

- Guest pays → the request enters a ****live queue the DJ sees****. Higher bids jump the queue; otherwise first-come.
- Hosts set a ****banned-song / do-not-play list****; anything filtered out is auto-refunded or rebooked to the next playable song.
- When played, a caption "Requested by [name]" flashes on the video wall or is announced.
- Money goes to the linked fund. Natural fit for bar/bat mitzvah kids and reception dance blocks.

4.3 Tier 2 – Guest Spotlight (drone talking head)

- ****Host side:**** fixed price per slot (\$50–150) or pay-what-you-want toward the fund; slot duration (30/45/60/90 s); bookable windows (cocktail hour, between courses, after first dance – never during vows/toasts); total slot cap (20–40).
- ****Guest side:**** books a live slot from the QR microsite ("7 of 40 left"), pays, gets a countdown. At slot time the pod lights ****ON AIR****, the nearest free drone swings to the seat's pre-mapped angle and records exactly the host-set duration – the guest never leaves their seat.
- ****The guarantee:**** the clip is tagged ****KEEP**** in the edit assembly; the final film is guaranteed to include it. That guarantee *is* the product.

4.4 Tier 3 – Video-wall message ("Big Screen")

The ****high-priced**** tier – the marquee "see yourself on the big screen" moment.

- Guest records a message up to the host-set cap (60–90 s) from their phone/webcam in the QR microsite, or in-venue into the pod camera. Remote guests can buy this too – a way to get on the wall without being there (ties into `§3.2`).
- Every message goes to a ****moderation queue****: hosts review and approve before it can air, to keep out anything inappropriate – embarrassing inside jokes, secrets, politics, hard language. Rejected messages get a quiet note and one re-record chance; no refund drama because nothing is final until hosts approve.
- Approved messages air on the video wall as ****interstitials**** between live segments (rotation), and stay eligible for a "best of" pull into the film if hosts opt in.
- ****Distinction vs. Spotlight:**** wall messages are **guaranteed to air** (if approved), but film inclusion is optional – Spotlight is the film guarantee, Big Screen is the live-air guarantee.

4.5 Tier 4 – Karaoke performance ("Stage")

The ****highest-priced**** tier.

- Guest buys a karaoke slot: picks from a ****licensed karaoke catalog****, books a window inside a host-approved karaoke/dance segment.
- At the slot, lyrics run on the video wall and audio goes through the PA; the drone (or fixed cam) covers the performance; the clip is tagged ****KEEP**** like a Spotlight – guaranteed in the film.
- ****Moderation:**** songs must be in the licensed catalog and host pre-approved; a live operator ****cut button**** kills audio/visuals if a performance goes sideways mid-song.
- Note: needs karaoke licensing handled (venue license or per-song), flagged in `§4.8`.

4.6 The booking rail (shared by all tiers)

- One **per-event QR link** → the event's microsite shows every enabled tier, live availability, prices and remaining spots.
- SKYCREW generates table-tent QR cards; the same link lives on the trigger pods.
- Payments (Stripe-class, Apple/Google Pay) clear instantly; each booking is a priority entry on the operator's timeline – color-coded by tier.

4.7 Orchestrator, moderation & the guarantee

- **Spotlight & karaoke** slots are **priority reservations**: the orchestrator pre-blocks a drone + fixed cam + PA for that window before the event starts, so paid moments are never skipped by a competing pod trigger. Paid shots win over ad-hoc focus requests.
- **Moderation dashboard**: every wall message, karaoke song and song request routes through one host-facing approval screen; nothing airs unapproved.
- **Film guarantee**: Spotlight and karaoke clips carry the **KEEP pin** – they cannot be starred-out in client review; if oversold they roll into a "Guest Spotlight" montage so every paid guest still appears.
- **Wall messages**: guaranteed to air if approved; film inclusion is a host opt-in.

4.8 Economics

- **Hosts**: the menu stacks. Example event: 30 Spotlight (\$75) + 8 wall messages (\$150) + 40 song requests (\$40) + 3 karaoke (\$300) ≈ **\$5,950** – enough to cover a large share of the whole film package, and the fund gets topped up by guests who were already going to give cash.
- **Partners / SKYCREW**: flat **\$49/event** unlock, or ~5–10% platform fee on paid items routed through the webapp's payment layer. Sits naturally in the partner revenue-share from ``03-partner-model.md``.
- **Guests**: a zero-effort gift that's also an appearance or a song – the emotional wedge that sells it.

4.9 Notes & risks

- **Scheduling** – bookable windows only; operator can override/move a slot live on the timeline.
- **Moderation liability** – hosts hold final approval; guests consent to reuse by paying; venue privacy signage unchanged (``00-overview.md §5``).
- **Song/karaoke rights** – licensed catalog only; DJ/PA integration requires the venue or DJ to cooperate (the operator brings the mixer feed hookup).
- **Queue abuse** – cap purchases per guest (e.g. 2–3 items); song requests auto-filtered by the banned list.
- **Conflicts** – one Spotlight/karaoke slot at a time by default; windows staggered ~5 min.
- **No-shows** – slot auto-releases after ~60 s grace, host notified, guest gets a refund or a makeup slot if the timeline allows.

Part 4 – In-venue sight-line feeds (big venues / bad sight lines)

Problem: in large halls, pillar-heavy rooms, balconies or long ballrooms, a big slice of the audience can't actually see the action. Sell them the live camera feed instead. This is the *venue-to-guest* counterpart of Part 3.1 (which is *venue-to-remote*), and mirrors how Resi sells "synchronized video across locations / overflow rooms" to venues.

Two delivery methods – customer picks either or both:

4.1 Table monitors (live feed at the tables)

- One display per table (or per two tables) showing the broadcast – either the operator's multiview or a chosen "hero angle" feed.
- **Zero-hardware option:** any tablet or smart TV at the table runs the live viewer as a Chromium webview. No extra hardware beyond the screen (many venues already own TVs).
- **HDMI option:** 24–32" panels (\$150–350 each) + optional HDMI-over-LAN decoder boxes (\$100–200 each) for venues that want real HDMI outputs on a house system.
- **House-system option:** feed the venue's existing projectors / video wall via SDI or HDMI for the crowd that can't see.

4.2 Sight-line webapp (guests' phones, venue LAN)

- The same private viewer link as remote guests (Part 3.1), but served **from the venue LAN** with zero-internet, low-latency streaming** so in-room guests with bad seats watch on their phones at ~1 s latency. No cellular data required, no dependency on venue uplink speed.

4.3 Tech: local-first streaming

- A **local WebRTC SFU (LiveKit)** runs on the operator's ground hub/laptop and distributes the live feeds over the venue Wi-Fi/LAN. Table monitors and phones are just webviews pointed at the local viewer.
- Remote guests keep the cloud path (Part 3.1). Same feed, two paths: **LAN for in-room, cloud for remote** – so a weak venue internet uplink never degrades the in-room experience.
- Latency budget: <2 s local, <5 s remote.
- If venue Wi-Fi is weak: mesh router add-on \$100–300 (BOM \$3).

4.4 Pricing (upsell)

Item	Price
---	---
Sight-Line Pack (any 2 delivery methods: table monitors + webapp)	***\$49–99/event**
Bundled free on Wedding Duo / Condor Pro with the streaming bundle (S3 stretch)	included
Table panels 24–32" (optional hardware, per unit)	\$150–350 each
HDMI-over-LAN decoder boxes (optional)	\$100–200 each
Mesh router (optional, weak venue Wi-Fi)	\$100–300

> Sales angle for operators: "Half your guests won't see the ceremony from those seats – give them the best seat in the house on the screen in front of them." Same mechanism as the remote-guest stream, so no new technology – pure margin.

Rollout order (tied to campaign stretch goals, `02-campaign.md`)

Milestone	When
---	---
Venue Planner alpha (S1 \$150K)	6–9 mo post-campaign
Venue Planner GA + Edit module (S3 \$350K)	9–12 mo
Live-stream bundle + in-venue sight-line feeds (S3 \$350K)	9–12 mo
Video-wall layouts	12–18 mo (revenue driver for franchise-lite)